

High School Contest Preparation

Date undecided

Contents

1	High School Math Brief	4
2	Introduction	4
3	Special Thanks	5
4	Chapter 1: Identities and Basic Factorization	6
4.1	Basic Identities	6
4.2	Difference of Squares	8
4.3	Sum and Difference of Cubes	10
4.4	Other Common Expansions	12
4.5	SFFT (Simon's Favorite Factoring Trick)	13
4.6	Sophie-Germain Identity	16
4.7	Challenge Problems	17
5	Chapter 2: Quadratics & Other Polynomials	19
5.1	Deriving The Quadratic Formula	19
5.2	Polynomial Division	21
5.3	The Rational Root Theorem	23
5.4	Vieta's Formulas for Quadratics	25
5.5	Graphing Polynomials	26
5.6	Challenge Problems	28
6	Chapter 3: Geometry Basics	30
6.1	Angles	30
6.1.1	Degrees	30
6.1.2	Radians	31

7	Chapter 4: Triangles	33
7.1	Pythagorean Theorem	33
7.2	Sum of Angles in a Triangle	34
7.3	Introducing Special Triangles	35
7.4	Area and Perimeter of a Triangle	37
7.5	Congruent Triangles	38
7.6	Similar Triangles	39
7.7	Solving Contest Problems with Similar Triangles	40
7.8	Special Parts of Triangles	42
7.9	Challenge Problems	43
8	Chapter 5: Polygons	44
8.1	Polygon Basics	44
8.2	Polygon Area and Perimeter	44
8.3	Solving Problems with Polygons	46
8.4	A Note Regarding Polygons	47
8.5	Challenge Problems	47
9	Chapter 6: 3D Figures	47
9.1	Introduction	47
9.2	Rectangular Prisms	48
9.3	Cubes	48
9.4	Spheres	49
9.5	Cones	51
9.6	Cylinders	53
9.7	Square Pyramids	54
9.8	Tetrahedrons	56
9.9	Challenge Problems	57
10	Chapter 7: Counting & Probability	57
10.1	Introduction & the Basics	57
10.2	Casework and Exclusive Events	61
10.3	Permutations	63
10.4	Combinations	64
10.5	Expected Value	65
10.6	Stars and Bars	68
10.7	Pascal's Triangle & Pascal's Identity	69
10.8	Hockey Stick Identity	69
10.9	Binomial Theorem	70
10.10	Basic Probability	70

10.11	Advanced Probability	71
10.12	Challenge Problems	72
11	Chapter 8: Number Theory	73
11.1	Introduction & the Basics	73
11.2	Division Theorem	74
11.3	Prime Factorizations	74
11.4	Divisibility Rules	75
11.5	Base Number Systems and Arithmetic with Bases	76
11.6	Introduction to Modulos	77
11.7	Euclidean Algorithm and Extended Version	79
11.8	Chinese Remainder Theorem	80
11.9	The Chicken-McNugget Theorem	84
11.10	Challenge Problems	87
12	Chapter 9: Introducing Functions	90
12.1	Transforming Functions	93
12.2	Graphing Functions	93
12.3	Inverse Functions	95
12.4	Problem Solving in Functions	99
12.5	Absolute Value Function	103
12.6	Logarithms	106
12.7	Interval Notation	107
12.8	Domain	108
12.9	Ranges	109
12.10	Challenge Problems	111
13	Chapter 10: Analytic Geometry	112
13.1	Introduction	112
13.2	Lines	112
13.3	Circles and their Applications	113
13.4	Coordinate Bashing	114
13.5	Challenge Problems	114
14	Chapter 11: Trigonometry	114
14.1	Basic Trigonometry	115
14.2	The Pythagorean Trigonometry Identity	119
14.3	The Unit Circle	122
14.4	Trigonometry Properties	122

15 Chapter 12: Sequences & Series	123
15.1 Introduction & The Basics	123
15.2 Summation Notation	124
15.3 Finite Geometric Series	127
15.4 Infinite Geometric Sequences	129
15.5 Arithmetico-Geometric Series	130
15.6 Famous Series	132
16 Chapter 13: Proofs	134
16.1 Without Loss of Generality	135
16.2 Contradiction	136
16.3 Essential Inequalities	137
17 Chapter 14: Polynomials	142